

Experiment-2 Write a python program to solve 8 queen problem

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queen's.py - C:/Users/prana/OneDrive/Desktop/queen's.py (3.13.14)
File Edit Format Run Options Window Help

N = 8
board = [-1] * N

def safe(row, col):
    for i in range(row):
        if board[i] == col:
            return False

        if abs(board[i] - col) == abs(i - row):
            return False

    return True

def solve(row):
    if row == N:
        return True

    for col in range(N):
        if safe(row, col):
            board[row] = col

            if solve(row + 1):
                return True

            board[row] = -1

    return False

solve(0)

print("8 Queens Solution:")

for row in range(N):
    for col in range(N):
        if board[row] == col:
            print("Q", end=" ")
        else:
            print(".", end=" ")
    print()
```

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IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/prana/OneDrive/Desktop/queen's.py =====
8 Queens Solution:
Q . . . . .
. . . . Q . .
. . . . . Q
. . . . Q . .
. . Q . . . .
. . . . . Q .
. Q . . . . .
. . . Q . . .
>>> |
```